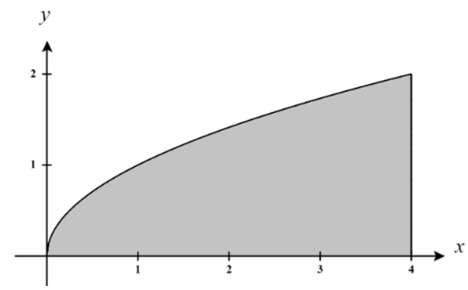
MTH 202 Chapter 4: Multiple Integrals

4.2 Double Integrals Worksheet

1. In the *xy*-plane, the region *R* is bounded by $y=\sqrt{x}$, $x=4$ and the *x*-axis. Determine the volume of the solid bounded above by $f(x,y)=\dfrac{y}{1+x^2}$ and below by the region *R*.



A picture of the region *R* is shown at the right. We see that the region *R* can be defined as $R=\{(x,y)|\ 0\le y\le\sqrt{x}\ \ \&\ \ 0\le x\le 4\}$ OR $R=\{(x,y)|\ y^2\le x\le 4\ \ \&\ \ 0\le y\le 2\}$

We know that the volume of the solid is given by

$$V=\int_0^4\int_0^{\sqrt{x}}\frac{y}{1+x^2}\,dydx \quad \text{OR} \quad V=\int_0^2\int_{y^2}^4\frac{y}{1+x^2}\,dxdy$$

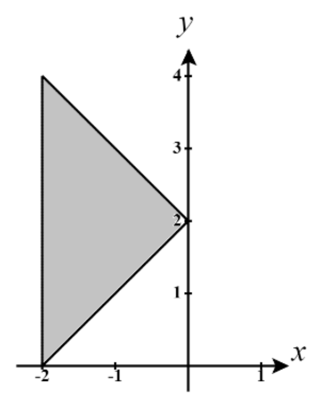
We will calculate the first double integral as follows:

$$V=\int_0^4\int_0^{\sqrt{x}}\frac{y}{1+x^2}\,dydx=\int_0^4\frac{1}{2}\frac{y^2}{1+x^2}\Bigg|_0^{\sqrt{x}}dx=\int_0^4\left[\frac{1}{2}\frac{x}{1+x^2}-\frac{1}{2}(0)\right]dx=\frac{1}{2}\int_0^4\frac{x}{1+x^2}\,dx$$

Let $u=1+x^2$ then $du=2xdx \ \rightarrow\ \frac{1}{2}du=xdx$. Our bounds become $u(0)=1$ & $u(4)=17$. Then,

$$\frac{1}{2}\int_0^4\frac{x}{1+x^2}\,dx=\frac{1}{4}\int_1^{17}\frac{1}{u}\,du=\frac{1}{4}\ln|u|\Bigg|_1^{17}=\frac{1}{4}\ln(17)-\frac{1}{4}\ln(1)=\frac{1}{4}\ln(17)$$

2. In the *xy*-plane, the region *R* is bounded by the curves $y=x+2$, $y=2-x$ and $x=-2$. Sketch the region and determine the volume of the solid bounded above by $f(x,y)=x^2$.



A picture of the region *R* is shown at the right. We see that the region *R* is most easily defined as $R=\{(x,y)|\ x+2\le y\le 2-x\ \ \&\ \ -2\le x\le 0\}$

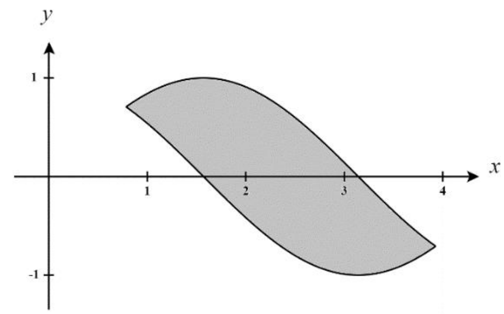
We know that the volume of the solid is given by $V=\int_{-2}^{0}\int_{x+2}^{2-x}x^2\,dydx$.

We will calculate the double integral as follows:

$$V=\int_{-2}^{0}\int_{x+2}^{2-x}x^2\,dydx=\int_{-2}^{0}x^2y\Big|_{x+2}^{2-x}dx=\int_{-2}^{0}\left[x^2(2-x)-x^2(x+2)\right]dx=\int_{-2}^{0}\left[-2x^3\right]dx$$

$$=-\frac{1}{2}x^4\Bigg|_{-2}^{0}=0-\left(-\frac{1}{2}(16)\right)=8$$

3. Use a double integral to compute the area of the region bounded by the graphs of $f(x)=\sin(x)$ and $g(x)=\cos(x)$ between $x=\frac{\pi}{4}$ and $x=\frac{5\pi}{4}$.



A picture of the region R is shown at the right. We see that the region R is most easily defined as

$$R=\left\{(x,y)\mid \cos(x)\le y\le \sin(x)\ \&\ \frac{\pi}{4}\le x\le \frac{5\pi}{4}\right\}$$

We know that the volume of the solid is given by $V=\int_{\pi/4}^{5\pi/4}\int_{\cos(x)}^{\sin(x)} 1\,dydx$.

We will calculate the double integral as follows:

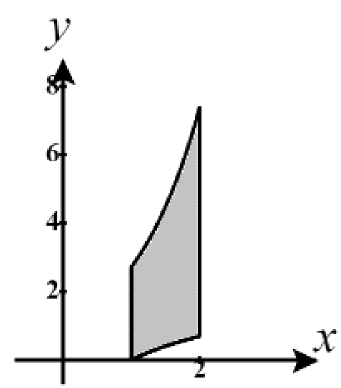
$$A=\int_{\pi/4}^{5\pi/4}\int_{\cos(x)}^{\sin(x)} 1\,dydx=\int_{\pi/4}^{5\pi/4}\left[\sin(x)-\cos(x)\right]dx=-\cos(x)-\sin(x)\Big|_{\pi/4}^{5\pi/4}$$

$$=-\cos\left(\frac{5\pi}{4}\right)-\sin\left(\frac{5\pi}{4}\right)+\cos\left(\frac{\pi}{4}\right)+\sin\left(\frac{\pi}{4}\right)$$

$$=\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}$$

$$=2\sqrt{2}$$

4. Use a double integral to determine the area of the region D bounded by $y=\ln(x)$, $y=e^x$ on the interval $1\le x\le 2$.



A picture of the region D is shown at the right. We see that the region D can be most easily defined as $D=\left\{(x,y)\mid \ln(x)\le y\le e^x\ \&\ 1\le x\le 2\right\}$.

We know that the area of the region is given by $V=\int_1^2\int_{\ln(x)}^{e^x} dydx$

We will calculate the first double integral as follows:

$$A=\int_1^2\int_{\ln(x)}^{e^x} dydx=\int_1^2\left(e^x-\ln(x)\right)dx=\int_1^2 e^x dx-\int_1^2 \ln(x)\,dx$$

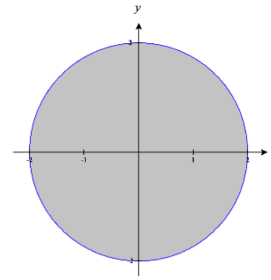
While the first integral here is easy, we will need to use integration by parts on the 2nd integral.

Let $u=\ln(x)$ and $dv=dx$ then $du=\frac{1}{x}$ & $v=x$. Then,

$$\int_1^2 e^x dx-\int_1^2 \ln(x)\,dx=e^2-e^1-\left[x\ln(x)\Big|_1^2-\int_1^2 x\frac{1}{x}dx\right]=e^2-e-\left[2\ln(2)-(2-1)\right]=e^2-e-2\ln(2)+1$$

5. Evaluate $\iint\limits_D (2x-y)\,dA$ where D is the region bounded by circle centered at the origin with radius 2.

A picture of the region D is shown at the right. We see that D can be defined as
$D=\left\{(x,y) \mid -\sqrt{4-x^2} \le y \le \sqrt{4-x^2} \ \ \& \ -2 \le x \le 2\right\}$



Therefore,

$$\iint\limits_D (2x-y)\,dA = \int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (2x-y)\,dy\,dx = \int_{-2}^{2} \left(2xy-\frac{1}{2}y^2\right)\Bigg|_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx$$

$$= \int_{-2}^{2}\left[\left(2x\sqrt{4-x^2}-\frac{1}{2}\left(4-x^2\right)\right)-\left(-2x\sqrt{4-x^2}-\frac{1}{2}\left(4-x^2\right)\right)\right]dx = \int_{-2}^{2} 4x\sqrt{4-x^2}\,dx$$

Let $u=4-x^2$, then $du=-2x\,dx \ \rightarrow\ -du=2x\,dx$. The bounds become $u(-2)=0$ & $u(2)=0$.

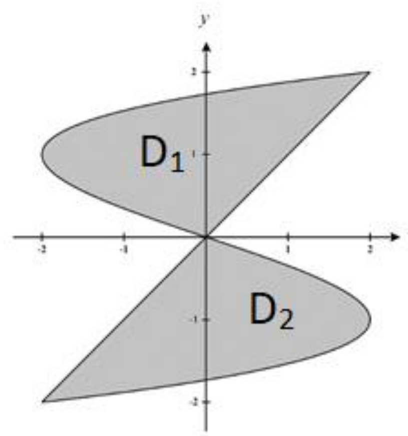
Therefore, $\displaystyle\int_{-2}^{2} 4x\sqrt{4-x^2}\,dx = \int_{0}^{0} -2\sqrt{u}\,du = 0$

6. Evaluate $\iint\limits_D y\,dA$ where D is the region bounded by $y^3-3y-x=0$ and $y=x$.

A picture of the region D is shown at the right. We see that the region D is difficult to define as a single type 1 or type 2 region. Therefore, we will split the region D up into two subregions D_1 and D_2 (upper and lower halves). We see that:

$D_1=\left\{(x,y) \mid 0 \le y \le 2 \ \ \& \ \ y^3-3y \le x \le y\right\}$ &

$D_2=\left\{(x,y) \mid -2 \le y \le 0 \ \ \& \ \ y \le x \le y^3-3y\right\}$



We will set up double integrals over D_1 and D_2 and their sum will be the total area we desire. We calculate:

$$\iint\limits_D y\,dA = \iint\limits_{D_1} y\,dA + \iint\limits_{D_2} y\,dA = \int_{0}^{2}\int_{y^3-3y}^{y} y\,dx\,dy + \int_{-2}^{0}\int_{y}^{y^3-3y} y\,dx\,dy$$

$$= \int_{0}^{2} xy\Big|_{y^3-3y}^{y} dy + \int_{-2}^{0} xy\Big|_{y}^{y^3-3y} dy$$

$$= \int_{0}^{2}\left(y^2-y^4+3y^2\right)dy + \int_{-2}^{0}\left(y^4-3y^2-y^2\right)dy$$

$$= \int_{0}^{2}\left(4y^2-y^4\right)dy + \int_{-2}^{0}\left(y^4-4y^2\right)dy$$

$$= \left(\frac{4}{3}y^3-\frac{1}{5}y^5\right)\Bigg|_{0}^{2} + \left(\frac{1}{5}y^5-\frac{4}{3}y^3\right)\Bigg|_{-2}^{0}$$

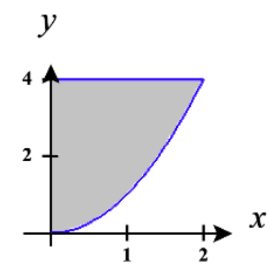
$$= \left(\frac{32}{3}-\frac{32}{5}\right)-(0)+(0)-\left(\frac{-32}{5}+\frac{32}{3}\right)=0$$

7. Sketch the region of integration and change the order of integration.

a) $\int_0^2 \int_{x^2}^4 f(x,y)\,dydx$

The bounds on the integral communicate the region
$D = \{(x,y) \mid 0 \le x \le 2 \ \&\ x^2 \le y \le 4\}$.
A sketch of the region is shown at the right.
We then redefine this region as $D = \{(x,y) \mid 0 \le x \le \sqrt{y} \ \&\ 0 \le y \le 4\}$.

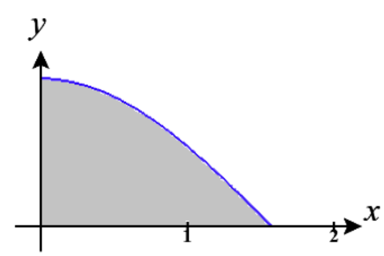
Therefore, $\int_0^2 \int_{x^2}^4 f(x,y)\,dydx = \int_0^4 \int_0^{\sqrt{y}} f(x,y)\,dxdy$.



b) $\int_0^{\pi/2} \int_0^{\cos(x)} f(x,y)\,dydx$

The bounds on the integral communicate the region
$D = \left\{(x,y) \mid 0 \le x \le \frac{\pi}{2} \ \&\ 0 \le y \le \cos(x)\right\}$.
A sketch of the region is shown at the right.
We then redefine this region as
$D = \{(x,y) \mid 0 \le x \le \arccos(y) \ \&\ 0 \le y \le 1\}$.

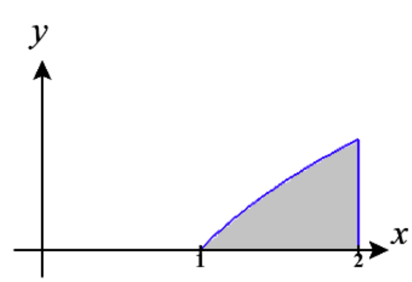
Therefore, $\int_0^{\pi/2} \int_0^{\cos(x)} f(x,y)\,dydx = \int_0^1 \int_0^{\arccos(y)} f(x,y)\,dxdy$.



c) $\int_1^2 \int_0^{\ln(x)} f(x,y)\,dydx$

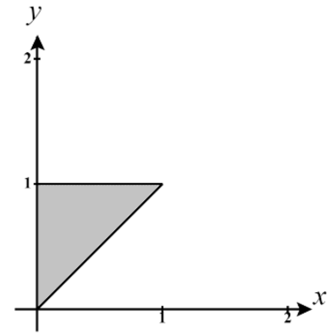
The bounds on the integral communicate the region
$D = \{(x,y) \mid 1 \le x \le 2 \ \&\ 0 \le y \le \ln(x)\}$.
A sketch of the region is shown at the right.
We then redefine this region as
$D = \{(x,y) \mid e^y \le x \le 2 \ \&\ 0 \le y \le \ln(2)\}$.

Therefore, $\int_1^2 \int_0^{\ln(x)} f(x,y)\,dydx = \int_0^{\ln(2)} \int_{e^y}^2 f(x,y)\,dxdy$.



8. Evaluate the double integral $\int_0^1 \int_x^1 e^{x/y} dy dx$.

We see that integrating with respect to y first will be difficult here. So, we will plan to reverse the order of integration. First, we must identify the region we are integrating over. We see that $D = \{(x,y) \mid 0 \le x \le 1 \text{ \& } x \le y \le 1\}$ and we picture this region at the right (Note that the equation of the slanted line is given by $y = x$). We re-express this region as: $D = \{(x,y) \mid 0 \le x \le y \text{ \& } 0 \le y \le 1\}$



Then,

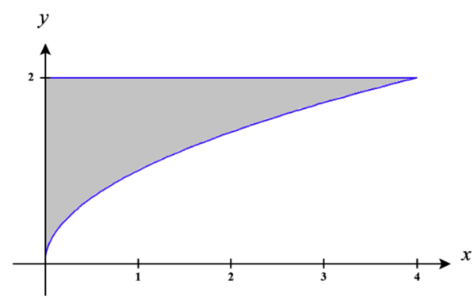
$$\int_0^1 \int_x^1 e^{x/y} dy dx = \int_0^1 \int_0^y e^{x/y} dx dy = \int_0^1 y e^{x/y} \Big|_0^y dy = \int_0^1 \left(y e^1 - y e^0 \right) dy = \int_0^1 y(e-1) dy$$

$$= \frac{1}{2} y^2 (e-1) \Big|_0^1$$

$$= \frac{1}{2}(e-1)$$

9. Evaluate the double integral $\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3+1} dy dx$.

We see that integrating with respect to y first will be difficult here. So, we will plan to reverse the order of integration. First, we must identify the region we are integrating over. We see that $D = \{(x,y) \mid 0 \le x \le 4 \text{ \& } \sqrt{x} \le y \le 2\}$ and we picture this region at the right. We re-express this region as: $D = \{(x,y) \mid 0 \le x \le y^2 \text{ \& } 0 \le y \le 2\}$



Then,

$$\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3+1} dy dx = \int_0^2 \int_0^{y^2} \frac{1}{y^3+1} dx dy = \int_0^2 \frac{x}{y^3+1} \Bigg|_0^{y^2} dy = \int_0^2 \frac{y^2}{y^3+1} dy$$

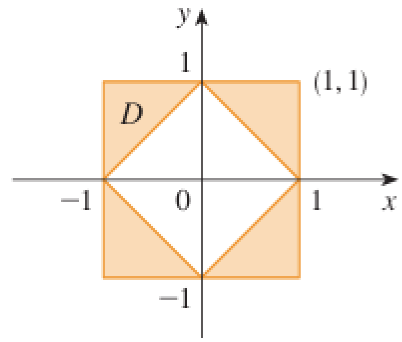
Let $u = y^3 + 1$, then $du = 3y^2 dy \to \frac{1}{3} du = y^2 dy$.

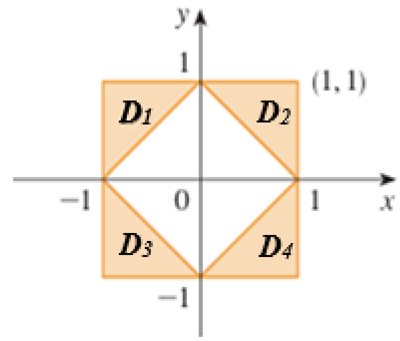
The bounds on the integral become $u(0) = (0)^3 + 1 = 1$ & $u(2) = (2)^3 + 1 = 9$.

Therefore, $\int_0^2 \frac{y^2}{y^3+1} dy = \int_1^9 \frac{1}{3} \cdot \frac{1}{u} du = \frac{1}{3} \ln|u| \Big|_1^9 = \frac{1}{3}\left(\ln(9) - \ln(1)\right) = \frac{1}{3}\ln(9)$.

10. Express D as a union of regions and evaluate the integral.

a) $\iint\limits_D x^2 dA$





We note that $\iint\limits_D x^2 dA = \iint\limits_{D_1} x^2 dA + \iint\limits_{D_2} x^2 dA + \iint\limits_{D_3} x^2 dA + \iint\limits_{D_4} x^2 dA$.

We define each of these regions in the following ways.

$D_1 = \{(x,y) \mid 1+x \le y \le 1 \ \& \ -1 \le x \le 0\}$

$D_2 = \{(x,y) \mid 1-x \le y \le 1 \ \& \ 0 \le x \le 1\}$

$D_3 = \{(x,y) \mid -1 \le y \le -1-x \ \& \ -1 \le x \le 0\}$

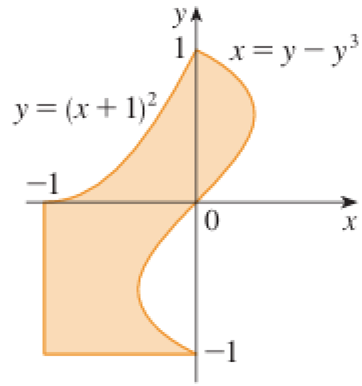
$D_4 = \{(x,y) \mid -1 \le y \le x-1 \ \& \ 0 \le x \le 1\}$

Therefore,

$$\begin{aligned}
\iint\limits_D x^2 dA &= \iint\limits_{D_1} x^2 dA + \iint\limits_{D_2} x^2 dA + \iint\limits_{D_3} x^2 dA + \iint\limits_{D_4} x^2 dA \\
&= \int_{-1}^{0}\int_{1+x}^{1} x^2 dy dx + \int_{0}^{1}\int_{1-x}^{1} x^2 dy dx + \int_{-1}^{0}\int_{-1}^{-1-x} x^2 dy dx + \int_{0}^{1}\int_{-1}^{x-1} x^2 dy dx \\
&= \int_{-1}^{0} x^2 y\Big|_{1+x}^{1} dx + \int_{0}^{1} x^2 y\Big|_{1-x}^{1} dx + \int_{-1}^{0} x^2 y\Big|_{-1}^{-1-x} dx + \int_{0}^{1} x^2 y\Big|_{-1}^{x-1} dx \\
&= \int_{-1}^{0}\left[x^2 - x^2(1+x)\right]dx + \int_{0}^{1}\left[x^2 - x^2(1-x)\right]dx + \int_{-1}^{0}\left[x^2(-1-x)+x^2\right]dx + \int_{0}^{1}\left[x^2(x-1)+x^2\right]dx \\
&= \int_{-1}^{0}\left[-x^3\right]dx + \int_{0}^{1}\left[x^3\right]dx + \int_{-1}^{0}\left[-x^3\right]dx + \int_{0}^{1}\left[x^3\right]dx \\
&= 2\int_{-1}^{0}\left[-x^3\right]dx + 2\int_{0}^{1}\left[x^3\right]dx \\
&= 2\left(-\frac{1}{4}x^4\right)\Bigg|_{-1}^{0} + 2\left(\frac{1}{4}x^4\right)\Bigg|_{0}^{1} \\
&= 2\left(0-\left(-\frac{1}{4}\right)\right) + 2\left(\frac{1}{4}-0\right) \\
&= 1
\end{aligned}$$

Note that we could use symmetry to help speed up this process. We have to be careful though. While $D_1, D_2, D_3, \& D_4$ are all symmetrical the volumes over these regions and under $f(x,y) = x^2$ are *NOT* all symmetrical. We have to consider not only the symmetry of the regions $D_1, D_2, D_3, \& D_4$, but the symmetry of the overall 3D regions that we are finding the volumes of. When we do this we can see that the volumes over $D_1 \ \& \ D_2$ are symmetrical and the volumes over $D_3 \ \& \ D_4$ are symmetrical. Therefore, we can say that $\iint\limits_D x^2 dA = 2\iint\limits_{D_1} x^2 dA + 2\iint\limits_{D_3} x^2 dA$ and shortcut the work here substantially.

b) $\iint\limits_D y\,dA$



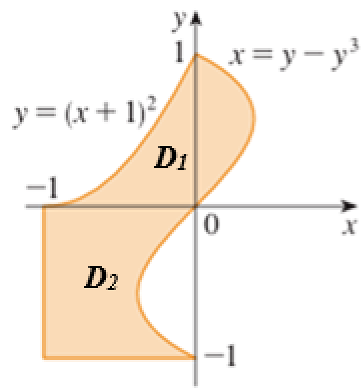
We break the region apart as shown in the image below. We then recognize that $\iint\limits_D y\,dA = \iint\limits_{D_1} y\,dA + \iint\limits_{D_2} y\,dA$.

We define each of these regions in the following ways.

$D_1 = \left\{(x,y) \mid \sqrt{y} - 1 \le x \le y - y^3 \ \& \ 0 \le y \le 1\right\}$

$D_2 = \left\{(x,y) \mid -1 \le x \le y - y^3 \ \& \ -1 \le y \le 0\right\}$

Therefore,



$$\iint\limits_D y\,dA = \iint\limits_{D_1} y\,dA + \iint\limits_{D_2} y\,dA = \int_0^1 \int_{\sqrt{y}-1}^{y-y^3} y\,dxdy + \int_{-1}^0 \int_{-1}^{y-y^3} y\,dxdy$$

$$= \int_0^1 \frac{1}{2}y^2\Big|_{\sqrt{y}-1}^{y-y^3} + \int_{-1}^0 \frac{1}{2}y^2\Big|_{-1}^{y-y^3}\,dy$$

$$= \int_0^1 \left[\frac{1}{2}\left(y-y^3\right)^2 - \frac{1}{2}\left(\sqrt{y}-1\right)^2\right]dy - \int_{-1}^0 \left[\frac{1}{2}\left(y-y^3\right)^2 - \frac{1}{2}(-1)^2\right]dy$$

$$= \int_0^1 \left[\frac{1}{2}\left(y^2 - 2y^4 + y^6\right) - \frac{1}{2}\left(y - 2\sqrt{y} + 1\right)\right]dy - \int_{-1}^0 \left[\frac{1}{2}\left(y^2 - 2y^4 + y^6\right) - \frac{1}{2}\right]dy$$

$$= \frac{1}{2}\int_0^1 \left[y^2 - 2y^4 + y^6 - y + 2\sqrt{y} - \frac{1}{2}\right]dy - \frac{1}{2}\int_{-1}^0 \left[y^2 - 2y^4 + y^6 - \frac{1}{2}\right]dy$$

$$= \frac{1}{2}\left[\frac{1}{3}y^3 - \frac{2}{5}y^5 + \frac{1}{7}y^7 - \frac{1}{2}y^2 + \frac{4}{3}y^{3/2} - \frac{1}{2}y\right]\Bigg|_0^1 - \frac{1}{2}\left[\frac{1}{3}y^3 - \frac{2}{5}y^5 + \frac{1}{7}y^7 - \frac{1}{2}y\right]\Bigg|_{-1}^0$$

$$= \frac{1}{2}\left[\frac{1}{3} - \frac{2}{5} + \frac{1}{7} - \frac{1}{2} + \frac{4}{3} - \frac{1}{2}\right] - \frac{1}{2}\left[-\left(-\frac{1}{3} + \frac{2}{5} - \frac{1}{7} + \frac{1}{2}\right)\right]$$

$$= \frac{1}{2}\left(\left[\frac{5}{3} - \frac{2}{5} + \frac{1}{7} - 1\right] - \left[\frac{1}{3} - \frac{2}{5} + \frac{1}{7} - \frac{1}{2}\right]\right)$$

$$= \frac{1}{2}\left(\frac{5}{3} - \frac{2}{5} + \frac{1}{7} - 1 - \frac{1}{3} + \frac{2}{5} - \frac{1}{7} + \frac{1}{2}\right)$$

$$= \frac{1}{2}\left(\frac{4}{3} - \frac{1}{2}\right)$$

$$= \frac{1}{2}\left(\frac{5}{6}\right)$$

$$= \frac{5}{12}$$